Lab 3 – Architecture Selection and Justification

Automated Rubric Assignment Evaluator

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Architecture Selection

We chose a Layered Architecture for the Automated Rubric Assignment Evaluator.It naturally divides system into three parts:Presentation Layer for Student and Faculty Evaluator screens,Business Layer for actual evaluation workflow, and Data Layer for submissions,test suites,rubrics,results and peer review assignments.This is good fit because the system has several connected tasks,but each task still has a clear purpose.

Reason 1 – Clear separation of evaluation responsibilities

Workflow is not just one operation.First a ZIP file has to be checked before it enters queue.Then configured test suite is run,results are converted into an itemized rubric score,and eligible submissions are assigned for peer review.Keeping these rules in separate business components makes flow easier to understand and test.For example,a Faculty Evaluator can update the marking rubric without affecting file validation or the student-facing portal.In the same way,peer-review assignment rule can be changed without changing already stored evaluation results.

Reason 2 – A practical fit for the lab scenario

Microservices would allow each service to be deployed and scaled separately,but that would add avoidable complexity for this lab:more services to manage,communication between services,possible consistency issues with queue and results.Basic client-server design is simpler,but it would put too much work into one central server and would not isolate test execution well.Layered architecture gives a practical middle ground. stay logically separate, while the system can still be built and deployed as one manageable application. If submissions grow, the Evaluation Worker can be increased as a worker

pool.

Security Advantage

Student and Faculty Evaluator portals do not access submission files or stored records directly.Every request first reaches the Business Layer,where login checks,role checks and input validation can be applied.Student can upload a project and see only their own score and peer-review assignment,while a Faculty Evaluator can access permitted submissions and configure test suites or rubrics.Evaluation Worker receives only a validated queued package and runs it in an isolated environment.This is especially important because submitted code must be treated as untrusted.

Performance Benefit

Submission Queue separates uploading from testing.So,if many students submit at same time,portal can acknowledge their uploads instead of making them wait for a test run to finish.Queued jobs can then be processed by Evaluation Workers and sent to the Rubric Scoring Service as structured test results.This helps the system retain work during the required 100 concurrent submissions and makes it possible to add worker capacity without changing the portal or database. It also helps completed results be displayed within the required 10 seconds.

How the Layers Work Together

Student uploads a ZIP through the portalSubmission Service validates it and puts it into the Submission Queue. Evaluation Worker picks up the job, runs the pre-configured unit tests and sends the result to the Rubric Scoring Service,which stores the itemized score.Peer Review Assignment Service assigns eligible submissions afterwards.When either actor wants to view information,request goes through the relevant business service rather than directly to the database.